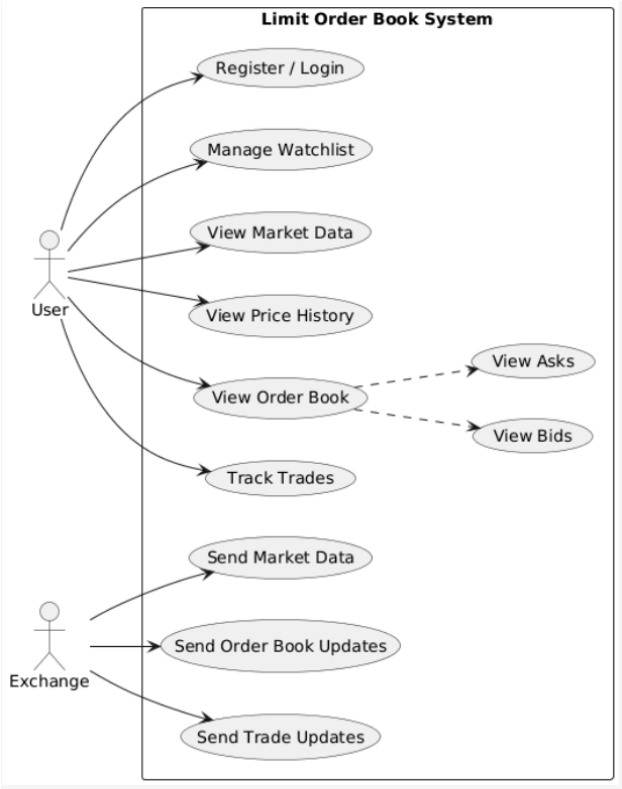
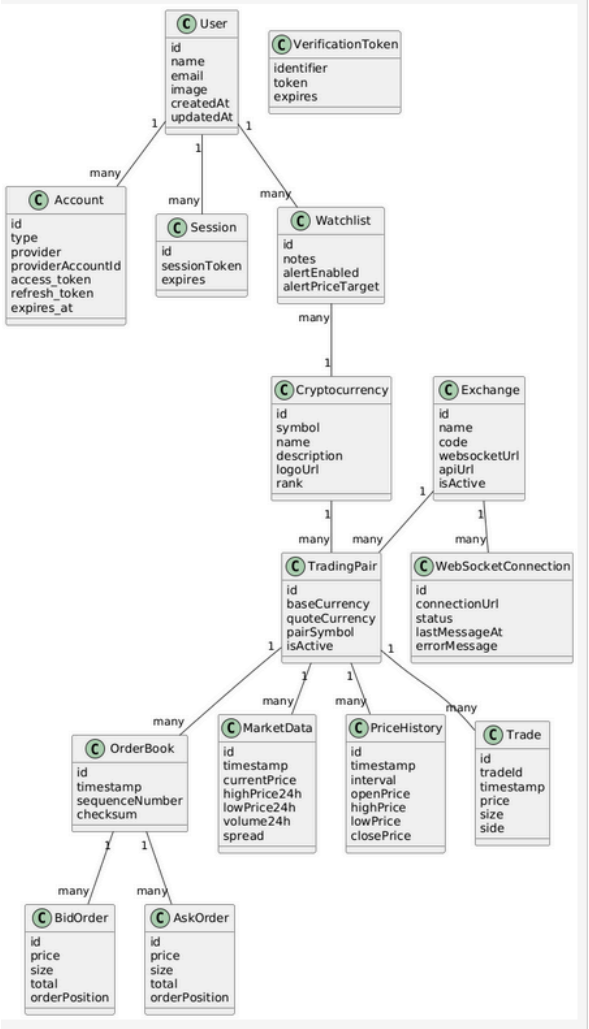
System Design using UML Diagrams

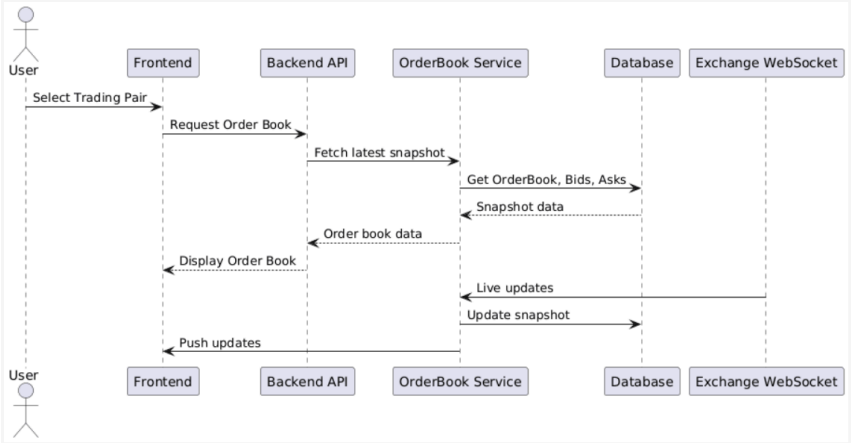
(1) Use case diagram



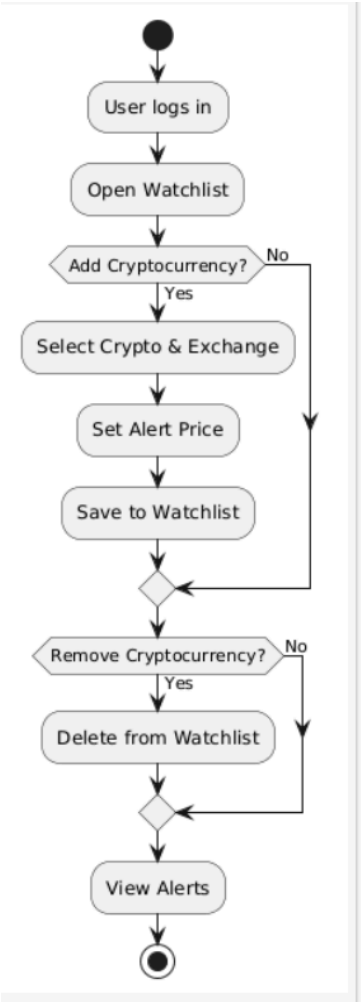
(2) Class diagram



(3) Sequence Diagram



(4) Activity Diagram



Description for Each Diagram

Diagram	Purpose	Module Represented	Elements
Use Case Diagram	Shows how external actors interact with the system and what major functionalities the system provides	System boundary of Limit Order Book platform	Actors: User, Exchange. Use cases: Login/Register, Manage Watchlist, View Market Data, View Order Book, Track Trades

Class Diagram	Represents structural design of the system by mapping entities to classes and showing associations	Core data model and object structure	Classes: User, Account, Session, Cryptocurrency, Exchange, TradingPair, OrderBook, BidOrder, AskOrder, MarketData, PriceHistory, Trade, Watchlist, WebSocketConnection. Relationships reflecting ER cardinalities
Sequence Diagram	Demonstrates runtime interaction when a user views the order book and how data flows through components	Order Book retrieval & live update workflow	Participants: User, Frontend, Backend API, OrderBook Service, Database, Exchange WebSocket. Messages showing request/response and live updates
Activity Diagram	Describes workflow of managing a watchlist step-by-step	Watchlist management module	Activities: Login, Add crypto, Set alerts, Remove crypto, View alerts. Decisions and flow control

Alignment of UML Models with ER Diagram

The UML models are derived directly from the ER model of the Limit Order Book System. The ER model is used to create the Class Diagram because each and every entity in the ER model has been replaced by a corresponding class in the Class Diagram. For example, User is replaced by the User class, and TradingPair is replaced by the TradingPair class, and OrderBook is replaced by the OrderBook class. The relationships defined between the classes in the ER model, such as User Account, TradingPair OrderBook, OrderBook Bid/AskOrder, and TradingPair Trade, are maintained in the Class Diagram as associations between the given classes. The Use Case Diagram defines the interaction between User and Exchange using the corresponding classes. The Sequence Diagram defines the real-time interaction between components using the corresponding classes in the ER model, such as OrderBook, BidOrder, AskOrder, and Trade. The Activity Diagram defines the interaction between the Watchlist class and the corresponding classes User and Cryptocurrency by maintaining the many-to-many relationship with the Watchlist class from the ER model.